Experiment 6

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Experiment 6: Total Cost with Discount and Tax

Aim:

To calculate the total cost of a customer's purchase after applying a discount and tax.

Algorithm:

1. Define item prices and quantities.
2. Calculate the total cost of all items.
3. Calculate the discount amount.
4. Subtract the discount from the subtotal.
5. Calculate tax on the discounted amount.
6. Add tax to obtain the final cost.

Python Code (IDLE):

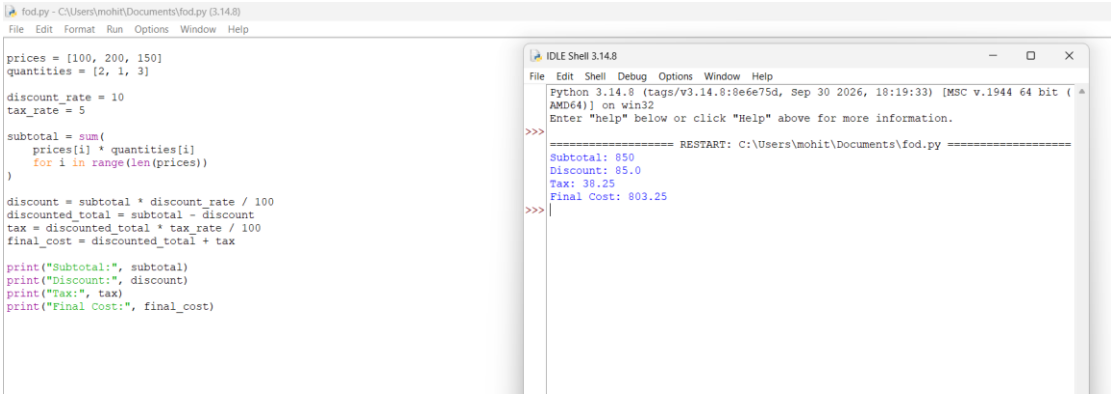
```
prices = [100, 200, 150]
quantities = [2, 1, 3]
discount_rate = 10
tax_rate = 5
subtotal = sum(    prices[i] * quantities[i]
for i in range(len(prices)))
discount = subtotal * discount_rate / 100
discounted_total = subtotal - discount
tax = discounted_total * tax_rate / 100
final_cost = discounted_total + tax
print("Subtotal:", subtotal)
print("Discount:", discount)
print("Tax:", tax)
print("Final Cost:", final_cost)
```

Input:

Prices = [100, 200, 150]
Quantities = [2, 1, 3]
Discount = 10%

Tax = 5%

Output:



Result:

Thus, the final purchase cost after applying a 10% discount and 5% tax was calculated as 803.25.